矩阵导数与MPC

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矩数的导数运算

标量对向量的导数

$\frac{df(y)}{dy}=0 \Rightarrow y^*$

推广

$\begin{cases} \frac{df(y_1,y_2)}{dy_1}=0 \\ \frac{df(y_1,y_2)}{dy_2}=0 \end{cases}$

$\Downarrow \vec{y}=\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$

$f(y_1,y_2)=f(\vec{y})$

定义 $\frac{\partial f(\vec{y})}{\partial(\vec{y})}=\begin{bmatrix} \frac{\partial f(\vec{y})}{\partial y_1} \\ \vdots \end{bmatrix}_{m\times 1}$

$\vec{y}$ 为 $m\times 1$

行数与分母相同时为Denominator layout

$1\times m$ 反之为Numerator layout

二者互为转置

向量方程对向量的导数

$\vec{f}(\vec{y})=\begin{bmatrix} f_1(\vec{y}) \\ \vdots \\ f_n(\vec{y}) \end{bmatrix}$ $\vec{y}=\begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}$

$\frac{\partial \vec{f}(\vec{y})}{\partial \vec{y}}=\begin{bmatrix} \frac{\partial f(\vec{y})}{\partial y_1} \\ \vdots \\ \frac{\partial f(\vec{y})}{\partial y_n} \end{bmatrix}$ 对 $\frac{\partial f(\vec{y})}{\partial y_1}$ 进行行子布局展开

分母布局

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$\frac{\partial A\vec{y}}{\partial \vec{y}} = A^T$ 分母布局结果

$\frac{\partial \vec{y}^T A \vec{y}}{\partial \vec{y}} = A\vec{y} + A^T\vec{y}$

$J = f[\vec{y}(\vec{u})]$, $\vec{y} = \begin{bmatrix} y_1(\vec{u}) \\ \vdots \\ y_n(\vec{u}) \end{bmatrix}$ $\vec{u} = \begin{bmatrix} u_1 \\ \vdots \end{bmatrix}$ $\vec{u}(n\times 1)$

$\frac{\partial J}{\partial \vec{u}} = \begin{bmatrix} \frac{\partial J}{\partial u_1} \\ \vdots \\ \frac{\partial J}{\partial u_n} \end{bmatrix}_{n\times 1}$

$\frac{\partial J}{\partial u_1} = \frac{\partial J}{\partial y_1(\vec{u})} \frac{\partial y_1(\vec{u})}{\partial u_1} + \cdots + \frac{\partial J}{\partial y_n(\vec{u})} \frac{\partial y_n(\vec{u})}{\partial u_n}$

故 $\frac{\partial J}{\partial \vec{u}} = \frac{\partial \vec{y}(\vec{u})}{\partial \vec{u}} \frac{\partial J}{\partial \vec{y}(\vec{u})}$

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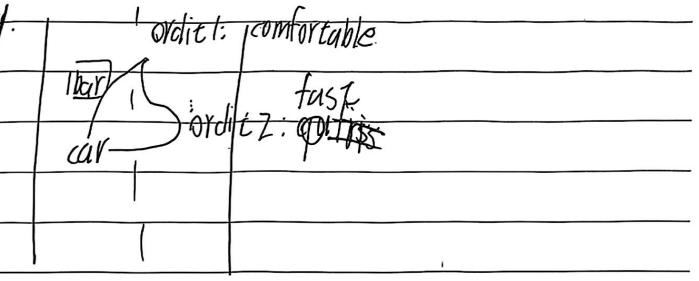
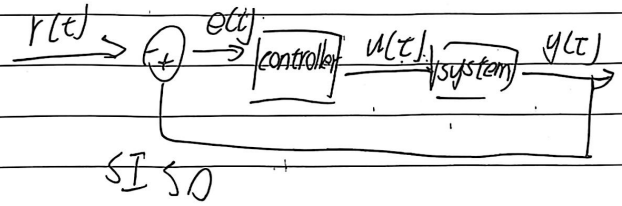
Optimal Control (MPC 部分)

Model Predictive Control:

~~Getting~~ the best performance whith Certain limitation

eg.





tracking: $\int_0^t e^2 dt \Rightarrow \downarrow$, track效果小

input: $\int_0^t u^2 dt \Rightarrow \downarrow$ ∴能耗↓

故可有 cost function: $J = \int_0^t q e^2 + r u^2 dt \Rightarrow \min J$

$q >> r$: 看重误差 $r >> q$: 看重输入

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推广到MIMO

$\frac{dx}{dt}=Ax+Bu$ 。x为state var

$Y=Cx$

$J=\int_0^{\infty} E^T Q E + u^T R u \, dt$

Q与R均为权重矩阵

MPC：通过模型来预测系统在某一未来时间段内的表现来进行最优化控制

多用于数位控制，故用离散化表示

$X_{k+1}=AX_k+BU_k$

step1：在k时刻估计/测量读取当前系统状态

step2：基于 $u_k, u_{k+1} \cdots u_{k+n}$ 来进行最优化

$J=\sum_{k=0}^{N-1} E_k^T Q E_k + U_k^T R U_k + E_N^T F E_N$

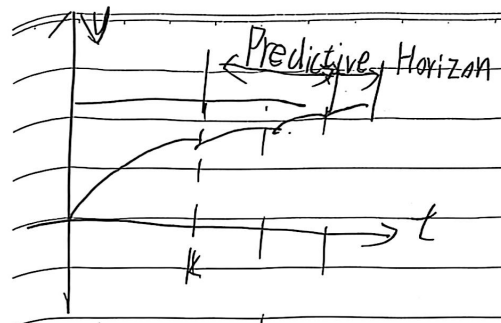
Terminal

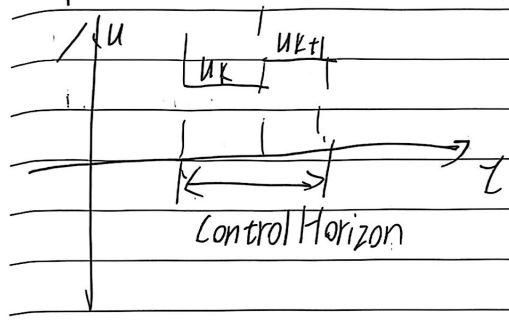
step3：只取 u_k

Receding Horizon Control

滚动优化控制

J考虑到了系统约束





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MPC推导:

step2: 二次规划 Quadratic ~~Program~~ Programming

$$\min J = \min\left(\frac{1}{2} z^T Q z + c^T z\right)$$

$$x_{k+1} = A x_k + B u_k$$

在k时刻: $\hat{U}_k = \begin{bmatrix} u(k|k) \\ u(k+1|k) \\ \vdots \\ u(k+N-1|k) \end{bmatrix}$ 和N为预测区间

$$\hat{X}_k = \begin{bmatrix} x(k|k) \\ \vdots \\ x(k+N|k) \end{bmatrix}$$

$$J = \sum_{i=0}^{N-1} \left(x(k+i|k)^T Q x(k+i|k) + u(k+i|k)^T R u(k+i|k) \right) + x(k+N)^T F x(k+N)$$

终端误差

终端误差的出现是基于x的反应慢于u一拍

$x(k|k) = x_k$

$$x(k+1|k) = A x(k|k) + B u(k|k) = A x_k + B u(k|k)$$

$$\vdots$$

$$x(k+N|k) = A^N x_k + A^{N-1} B u(k|k) + \cdots + B u(k+N-1|k)$$

即 $\hat{X}_k = \begin{bmatrix} I \\ A \\ \vdots \\ A^N \end{bmatrix} x_k + \begin{bmatrix} 0 & 0 & \cdots & 0 \\ B & 0 & \cdots & 0 \\ AB & B & & \\ \vdots & & \ddots & \\ A^{N-1}B & A^{N-2}B & \cdots & B \end{bmatrix} \hat{U}_k$ n个0行

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$$J = \hat{X}_k^T \begin{bmatrix} Q & & & \\ & Q & & \\ & & \ddots & \\ & & & Q_F \end{bmatrix} \hat{X}_k + \hat{U}_k \begin{bmatrix} R & & \\ & \ddots & \\ & & R \end{bmatrix}$$

则 $J = \hat{X}_k^T \bar{Q} \hat{X}_k + \hat{U}_k^T \bar{R} \hat{U}_k$

$$= (M\hat{x}_k + C\hat{u}_k)^T \bar{Q} (M\hat{x}_k + C\hat{u}_k) + \hat{u}_k^T \bar{R} \hat{u}_k$$

$$= \hat{x}_k^T M^T \bar{Q} M \hat{x}_k + 2\hat{x}_k^T C^T \bar{Q} M \hat{u}_k + \hat{u}_k^T C^T \bar{Q} C + \bar{R} \hat{u}_k$$

$$= \hat{x}_k^T G \hat{x}_k + 2\hat{x}_k^T E \hat{u}_k + \hat{u}_k^T H \hat{u}_k$$

$x_k^T G x_k$ 为const 此时 z^T 为 $\hat{u}_k$, $c^T = 2x^T E$

符合 $\min(z^T Q z + c^T z)$ 条件

上述为 $x \to 0$ 的建立, 建立了tracking的可用 $\hat{o} = \hat{x}_k - x$ 代替

通过使每一时刻的J达到最小, 从而校实现最优化